SAFETY & REGULATORY REVIEW

Pregnancy Support Belt with Eight FSR Sensors

Project design basis

A semi-automatic pregnancy support belt using eight force-sensitive resistors (FSRs), arranged as two sensors in each of four abdominal quadrants, to detect uneven loading or possible incorrect positioning and guide the wearer to make a manual adjustment.

Important scope note: This is a preliminary engineering and regulatory review for an academic design. It is not a clinical safety determination, certification, regulatory approval, or permission to test on pregnant volunteers.

Prepared as a design-review document

Regulatory context: India • Reference date: 24 October 2026

1. Executive Summary

The proposed belt combines a physical support garment with a pressure-sensing system. Its key safety principle is that it must not apply uncontrolled abdominal pressure or encourage excessive tightening. Sensor readings alone cannot establish that a pressure level is safe for a pregnant wearer or fetus.

The current concept is semi-automatic: the electronics detect sensor patterns and provide an indication or recommendation, while the wearer adjusts the belt manually. The design should not automatically tighten the belt.

Priority actions

- Define the exact intended purpose and intended user before selecting a regulatory classification.
- Complete a documented hazard analysis and risk-management file.
- Calibrate all eight sensors in the assembled belt and validate the full measurement chain.
- Test the mechanical design and electronics on suitable bench fixtures or a mannequin before any human testing.
- Obtain appropriate obstetric, biomedical-engineering, ethics and regulatory review before clinical use or commercial claims.

2. Design Description and Assumptions

Design assumptions used in this review

- Eight FSR sensors total: two in each of four abdominal quadrants.
- The system detects uneven sensor readings and may alert the user to check fit or position.
- The belt is semi-automatic and does not autonomously tighten itself.
- The electronics may include a microcontroller, signal-conditioning circuitry, voltage regulation, filtering capacitors, a power source and user feedback.
- The belt is intended to redistribute support pressure; its clinical effectiveness and pregnancy-specific safety have not yet been established.

These assumptions should be revised if the product is intended to measure abdominal pressure for clinical decisions, diagnose a condition, monitor fetal or maternal status, or claim to prevent pregnancy complications.

3. Indian Regulatory Review

The starting point is India's Medical Devices Rules, 2017, administered by the Central Drugs Standard Control Organisation (CDSCO). Classification and obligations depend on the product's intended purpose, claims, design and applicable classification entries—not merely on whether it contains electronics.

Regulatory position: Do not assume a device class or licensing pathway without checking the applicable CDSCO classification and requirements. A general support garment, a pressure-sensing aid and a device making clinical measurement claims may be treated differently. Seek qualified regulatory advice where the classification is uncertain.

Academic prototype

Engineering verification and bench testing may be possible within an academic project, subject to institutional rules and appropriate safety oversight. Prototype status does not mean the device is approved for clinical use or commercial sale.

Before a study involving pregnant participants

Obtain an appropriate protocol review, ethics approval where applicable, informed consent, qualified clinical oversight and any required regulatory permissions. Do not begin with pregnant volunteers before the bench-level design and measurement system have been reviewed and validated.

Before commercial sale

Confirm classification, licensing or registration requirements, applicable quality-system obligations, technical documentation, labelling, post-market obligations where applicable, and evidence supporting all safety and performance claims.

Useful official starting points

CDSCO Medical Devices:
https://www.cdsco.gov.in/opencms/opencms/en/Medical-Device-Diagnostics/Medical-Device-Diagnostics/

CDSCO obstetrical and gynecological device classification notice (verify current applicability): official notice PDF

4. Preliminary Hazard and Risk Register

The following register identifies foreseeable hazards and proposed controls. It is qualitative and preliminary; the project team must assign severity, probability, risk acceptability and verification evidence using a documented risk-management method.

Hazard	Possible consequence	Proposed risk controls
Excessive belt tension	Discomfort, skin pressure or potentially harmful abdominal compression	Adjustable mechanical design; clear fit instructions; clinician-reviewed design; easy release.
Uneven pressure distribution	Local pressure concentration despite apparently correct fit	Test all eight locations and the complete support structure under representative loading.
Incorrect FSR readings	False reassurance or unnecessary adjustment	Individual calibration; plausibility checks; explain measurement limitations.
Sensor failure or disconnection	Missed pressure hotspot	Detect open circuits, short circuits and implausible readings; show fault indication.
Incorrect positioning	Support applied in unintended locations	Clear orientation marks, fitting guidance and tests for misplacement.
Slipping during movement	Pressure shifts or reduced support effectiveness	Test retention and fit during sitting, standing, walking and bending on an appropriate fixture.
Skin irritation or heat	Redness, discomfort or contact reaction	Review contact materials, ventilation, cleaning and skin-contact duration.
Battery/electrical fault	Heating, leakage, short circuit or electrical injury	Protected low-voltage supply, insulation, enclosure and battery protection.
False or missed warning	User ignores warnings or trusts an unsafe fit	Validate thresholds, fault states, user messages and instructions.
Tightness when sitting	Increased discomfort or unwanted compression	Evaluate posture changes; ensure the belt can be loosened or removed promptly.

FSR limitation: A high sensor reading does not automatically mean pressure is dangerous, and a low reading does not prove the belt is safe. Calibration must cover the assembled belt and relevant contact, loading and movement conditions. FSR output should not be treated as validated clinical pressure unless the full measurement system has been validated for that purpose.

5. Technical Safety Requirements

A. FSR sensors and measurement

- Calibrate each of the eight sensors individually against a suitable reference measurement method.
- Characterise repeatability, drift, hysteresis, loading history, contact area and bending effects.
- Check sensor placement and mechanical load transfer; the sensor's output depends on how force is applied.
- Detect disconnected, damaged, saturated or implausible sensor signals.
- Avoid presenting a medical pressure value or a universal safe-pressure threshold without appropriate validation.

B. Mechanical design and materials

- Use soft, breathable, suitable skin-contact fabric and cover sharp edges and hard parts.
- Place electronics and hard components away from direct abdominal pressure points where practicable.

- Make the belt easy to loosen or remove quickly without assistance.
- Assess stitching, fasteners, slippage, fatigue, repeated fastening and cleaning effects.
- Review fit across intended body sizes and relevant postures; do not assume one geometry fits every pregnancy.

C. Electronics and firmware

- Verify regulator output, filtering, component temperature and electrical protection.
- Consider reverse-polarity, overcurrent and short-circuit protection as appropriate to the selected circuit.
- Ensure a sensor fault cannot be interpreted as normal pressure.
- Test reset, power loss, brownout and restart behavior; warnings must not silently remain disabled.
- Provide clear fault indication and a way to switch the electronics off.

D. Battery and charging

- Use a reputable, protected battery pack and a suitable charging circuit.
- Prevent sweat and cleaning fluid from reaching exposed circuitry.
- Evaluate abnormal heating, short circuit, overcurrent and foreseeable misuse.
- For rechargeable lithium batteries, assess applicability of IEC 62133-2 and the corresponding Indian standard, IS 16046 (Part 2); verify the current edition and any mandatory requirements.

6. Standards and Guidance to Assess

These are candidate references, not a declaration of compliance. Confirm the current editions, applicability, national adoption and regulatory requirements for the final intended purpose.

Reference	Potential relevance
ISO 14971:2019	Medical-device risk-management process: identify hazards, evaluate risks, implement controls and assess residual risk.
ISO 10993-1	Biological evaluation of materials that contact the body. The required evaluation depends on contact type, duration and risk.
IEC 60601-1	Consider if the product and intended use fall within the scope of medical electrical equipment requirements.
IEC 60601-1-11	Consider if the product qualifies as medical electrical equipment intended for use in the home healthcare environment.
IEC 62133-2 / IS 16046 (Part 2)	Potential battery safety references for portable sealed lithium cells and batteries; confirm exact applicability.

Do not claim ISO or IEC compliance merely because a standard is listed in the project report. Compliance requires assessment against applicable requirements and supporting evidence.

7. Verification and Validation Plan

Define acceptance criteria before testing. Record the method, equipment, sample count, conditions, results, deviations and pass/fail rationale. Do not use human subjects to discover basic mechanical or electrical failures.

Mechanical tests

- Inspect seams, fasteners, sharp edges, enclosure integrity and the ease of release.
- Evaluate slipping, repeated fastening, loosening and wear.
- Use a mannequin or suitable fixture to assess fit and pressure transfer in different postures.

Sensor tests

- Calibrate all eight sensors against a reference measurement system.
- Measure repeatability, drift and response under representative loading and contact conditions.
- Simulate open circuits, short circuits, disconnection and implausible readings.

Electronics tests

- Verify supply rails, regulator operation, component temperature and filtering.
- Test battery protection and relevant abnormal-power conditions.
- Check warning indications, restart behavior, power-loss recovery and safe shutdown.

System tests

- Test correct and incorrect belt placement on a mannequin or fixture.
- Test uneven loading and sensor-fault scenarios; confirm the display does not give false reassurance.
- Verify there is no automatic tightening or uncontrolled force adjustment.
- Assess foreseeable misuse, including overtightening, incorrect orientation and cleaning errors.

Documentation tests

- Maintain a hazard analysis and traceability table linking each risk control to a verification test.
- Keep calibration records, test results, software/firmware version and component specifications.
- Review instructions, warnings, cleaning and maintenance procedures.

Human testing

Any study involving pregnant participants requires a justified protocol, appropriate clinical and ethics review, informed consent and applicable regulatory approvals. Obtain expert review of the design and test evidence first. This document does not authorize human testing.

8. Labelling and User Instructions

- State intended purpose, intended user and important limitations.
- Explain correct orientation, fitting, adjustment and how to interpret each indicator.
- Explain how to loosen or remove the belt promptly.
- Specify cleaning instructions and whether the electronic module must be detached.
- Include battery charging and fault-handling instructions.
- Explain that FSR measurements are estimates of sensor response unless the complete system is validated for pressure measurement.

- Do not claim that the product prevents pregnancy complications or replaces antenatal care without adequate supporting evidence.

The instructions should advise the wearer to stop use if the belt causes pain, significant discomfort, skin irritation or breathing restriction and to seek appropriate medical advice. Pregnancy-specific warning signs must be reviewed by a qualified clinician before being included in user instructions.

9. Overall Preliminary Conclusion

Features worth retaining: eight-point sensing, manual adjustment, sensor-fault detection, easy release and a mechanically limited support design.

Main unresolved questions: whether the pressure distribution is appropriate for pregnancy, whether the FSRs produce reliable measurements in the assembled belt, and which CDSCO classification and standards apply to the final intended purpose.

Recommended next step: create a formal risk-management file, test protocol, acceptance criteria and regulatory classification rationale before testing the belt on people.

10. References and Further Review

1. CDSCO, Medical Devices portal: official website.

2. CDSCO, Obstetrical and Gynecological device classification notice (dated 03 June 2022; verify current applicability): official PDF.

3. ISO 14971:2019, Medical devices — Application of risk management to medical devices: ISO standard page.

4. ISO 10993-1, Biological evaluation of medical devices — Part 1: Evaluation and testing within a risk management process: ISO standard page.

Note: Confirm current standards editions and Indian regulatory requirements with the relevant standards body, CDSCO guidance and a qualified regulatory professional. Some standards may require purchase to access the complete normative text.

Disclaimer

This report is an initial educational engineering review based on the described concept. It is not legal, regulatory, obstetric or clinical advice, and it does not certify the design as safe, effective or compliant. Final decisions require review of the actual design, intended claims, materials, circuitry, software and test evidence by qualified professionals.